

ITEP 414 – System Administration and Maintenance

Week 2 Portfolio Project: Enterprise Infrastructure Planning for a Startup Company

Student Name: Delgado, Allyn Jade L.
Student ID: BSIT 4B
Program: Bachelor of Science in Information Technology
Course: ITEP 414 – System Administration and Maintenance
Week: 2
Company: ABC Startup Solutions
Project Type: Individual Portfolio Project

PART 1 – Company Profile

1.1 Company Name

ABC Startup Solutions

1.2 Nature of Business

ABC Startup Solutions is a new software and IT company that provides software development, web development, database solutions, IT consulting, and technical support for small and medium-sized businesses.

The company creates and maintains software for clients. It also provides technical support, system integration, and basic cloud services.

1.3 Company Overview

Field	Details
Company Name	ABC Startup Solutions
Office Location	4th Floor, TechHub Building, Greenfield Business Park, Los Baños, Laguna, Philippines (fictional)
Company Size	20 employees in a single office floor
Date Established	January 2026 (fictional)
Main Services	Custom software development, web and mobile applications, database solutions, IT consulting, and cloud-based business systems

1.4 Company Vision

To become a leading software solutions provider in the Philippines and Southeast Asia, recognized for delivering reliable, secure, and innovative technology that helps businesses grow in the digital economy.

1.5 Company Mission

ABC Startup Solutions is committed to delivering high-quality, scalable, and secure software solutions through continuous learning, technical excellence, strong client partnerships, and a well-managed IT infrastructure that supports sustainable business growth.

1.6 Organizational Structure

ABC Startup Solutions uses a flat and functional organizational structure suitable for a 20-person startup. The Chief Executive Officer oversees four department heads, while the IT Manager supervises the IT team and the Junior System Administrator handles infrastructure planning, deployment, and day-to-day technical support.

- Chief Executive Officer (CEO)
- IT Manager → IT Department (Junior System Administrator, Developers, Helpdesk staff)
- HR Manager → Human Resources Department
- Finance Manager → Finance Department
- Sales Manager → Sales Department

1.7 Employee Distribution

Department	Employees	Primary Function
Information Technology	5	Software development, infrastructure, and technical support
Human Resources	4	Recruitment, employee relations, and payroll coordination
Finance	5	Budgeting, billing, and financial reporting
Sales	6	Client acquisition, account management, and business development
TOTAL	20	—

PART 2 – Enterprise Hardware Inventory

The hardware inventory covers employee workstations and the shared infrastructure equipment required for daily operations. Desktop computers are assigned to employees who mainly work at fixed stations, while laptops are provided to IT and Sales staff who need mobility.

Asset ID	Hardware	Quantity	Department	Purpose
HW-001	Desktop Computer	9	HR (4), Finance (5)	Office productivity, HR records, and financial processing
HW-002	Laptop	11	IT (5), Sales (6)	Portable development work and client visits/presentations

HW-003	Server (Tower)	1	IT / Shared	Hosts internal applications, file sharing, Git repository, and Ubuntu Server workloads
HW-004	Router	1	Shared	Routes traffic between the LAN and ISP connection
HW-005	Switch (24-port)	1	Shared	Provides wired connectivity for workstations and network devices
HW-006	Printer	2	Shared	Network printing for documents, invoices, and reports
HW-007	UPS (1500VA)	3	Server Room / Network Rack / Finance	Protects equipment from power interruptions
HW-008	Wireless Access Point	2	Shared	Provides wireless coverage across the office floor
HW-009	NAS Storage (4-Bay, 8TB)	1	IT / Shared	Centralized file storage and scheduled backup target
HW-010	External Backup Drive (4TB)	2	IT / Offsite Rotation	Weekly offsite backup rotation for disaster recovery
HW-011	Monitor (24-inch)	20	All Departments	Display unit for every desktop and laptop user

2.1 Justification of Hardware Selections

- Desktops are assigned to HR and Finance because these roles mainly work from fixed desks and handle sensitive records.
- Laptops are assigned to IT and Sales because these employees benefit from portability for development, testing, client visits, and presentations.
- One physical server is appropriate for the current 20-user environment and can be virtualized later as the company grows.
- Two printers reduce downtime risk and make printing more convenient for different office areas.
- Three UPS units protect the server, network rack, and finance workstations from power interruptions.
- A separate NAS provides an additional backup location so data is not dependent on the main server.

PART 3 – Enterprise Software Inventory

The company uses a mix of free and licensed software for development, office work, and security.

Software	Version	License	Purpose
Windows 11 Pro	23H2	Volume License	Primary operating system for non-IT staff
Ubuntu Server	22.04 LTS	Free / Open Source	Operating system for the internal application and file server
Microsoft 365 Business Standard	Subscription	Per User	Word processing, spreadsheets, email, and presentations
VS Code	1.9x (latest)	Free	Primary code editor for the IT/development team
Git	2.44	Free / Open Source	Source control for software projects
GitHub Desktop	3.3	Free	Graphical interface for Git repositories
VirtualBox	7.0	Free / Open Source	Virtual machines for testing, staging, and training
Google Chrome	Latest stable	Free	Standard web browser for employees
Microsoft Defender	Built-in	Included	Baseline antivirus and endpoint protection
AnyDesk	Latest	Business License	Remote support and troubleshooting
7-Zip	23.01	Free / Open Source	File compression and archive management

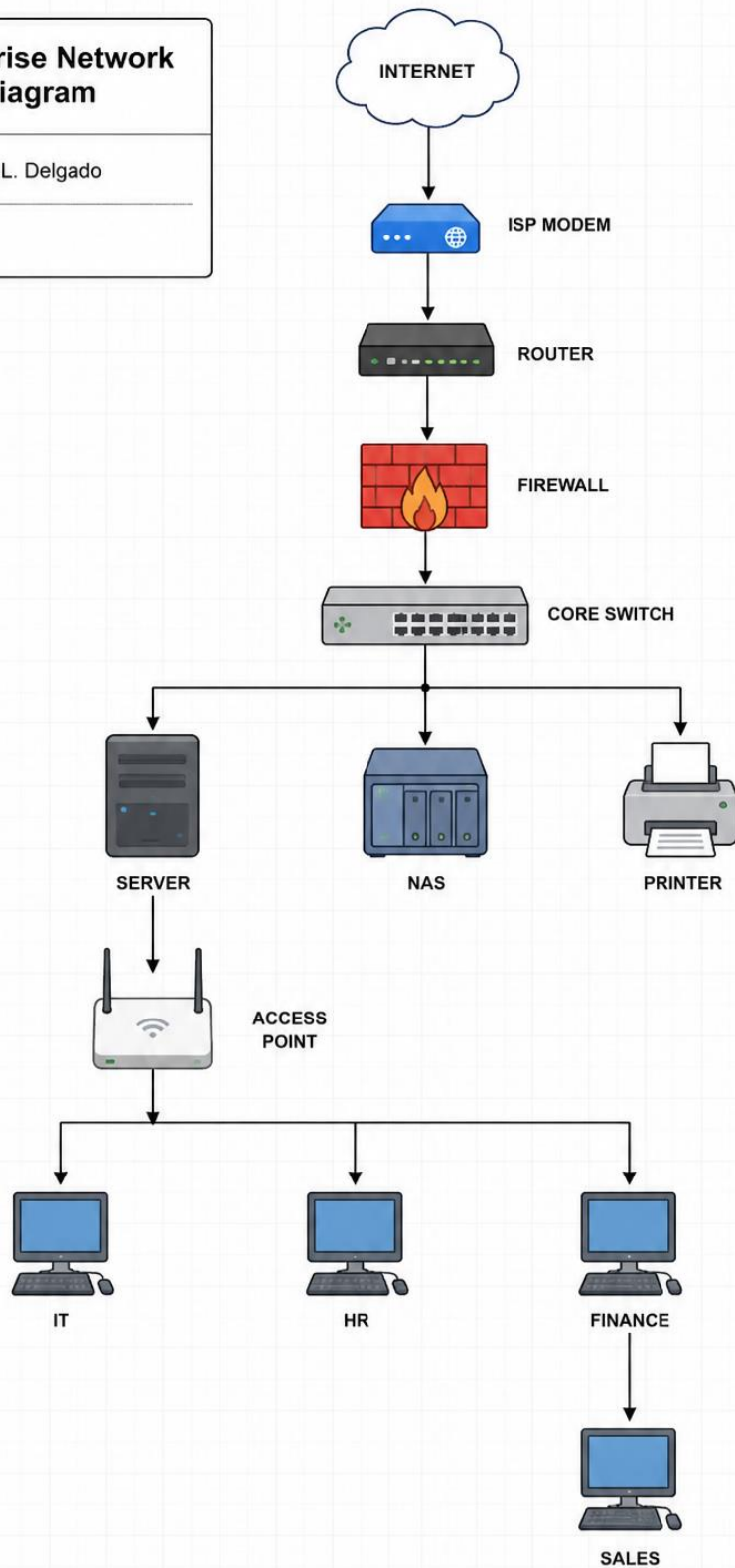
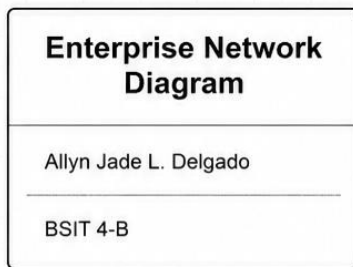
PART 4 – Enterprise Network Inventory

The network equipment is planned for a one-floor office with 20 users. Wired connections are used for desktop computers, while Wi-Fi is provided for laptops and mobile devices.

Asset ID	Equipment	Quantity	Specification	Purpose
NW-001	ISP Modem	1	Fiber modem, up to 300 Mbps	Terminates the ISP fiber line
NW-002	Router	1	Dual-WAN capable, Gigabit ports	Provides routing, NAT, and DHCP
NW-003	Firewall	1	UTM firewall appliance	Filters traffic, provides VPN termination, and

				intrusion prevention
NW-004	Switch (Core, 24-port)	1	Gigabit managed switch	Central wired connectivity point
NW-005	Wireless Access Point	2	Wi-Fi 6, dual-band	Wireless coverage for laptops and mobile devices
NW-006	Patch Panel (24-port)	1	CAT6 rated	Organizes and terminates network cabling
NW-007	CAT6 Cable	300 m	23 AWG, UTP	Structured cabling from patch panel to outlets
NW-008	RJ45 Connector	100 pcs	CAT6 rated	Cable termination and patch cords

PART 5 – Enterprise Network Diagram



5.1 Logical Traffic Flow

- Internet → ISP Modem
- ISP Modem → Router
- Router → Firewall
- Firewall → Core Switch
- Core Switch → Patch Panel / Server / Printers / Wireless Access Points
- Network connections → HR, Finance, IT, and Sales workstations

PART 6 – System Administration Roles

6.1 Helpdesk Technician

Responsibilities: Serves as the first point of contact for employee technical problems, records and resolves hardware, software, account, and connectivity issues, escalates complex problems, and performs basic hardware setup.

Skills: Troubleshooting, customer service, basic networking, operating system fundamentals, and ticketing workflows.

Common Tools: Freshservice, Zendesk, AnyDesk, TeamViewer, and basic Active Directory tools.

Certifications: CompTIA A+ and ITIL Foundation.

6.2 Network Administrator

Responsibilities: Designs, configures, and maintains routers, switches, firewalls, and wireless access points; monitors network performance; manages IP addressing, VLANs, VPN access, and network security policies.

Skills: Routing and switching, subnetting, firewall configuration, network monitoring, and troubleshooting.

Common Tools: Wireshark, PRTG/Zabbix, Cisco Packet Tracer, and pfSense/firewall consoles.

Certifications: CompTIA Network+ and Cisco CCNA.

6.3 Linux System Administrator

Responsibilities: Installs, configures, and maintains Linux servers; manages users, permissions, and services; automates routine tasks; monitors logs and system performance; and applies security updates.

Skills: Linux command line, Bash scripting, service and process management, package management, and backups.

Common Tools: SSH, cron, systemd, Bash, and basic Ansible automation.

Certifications: CompTIA Linux+ and RHCSA.

6.4 Cloud Administrator

Responsibilities: Manages cloud compute, storage, and networking resources; monitors usage and costs; implements cloud backup and disaster recovery; and manages identity and access.

Skills: IaaS/PaaS fundamentals, virtual networking, identity and access management, and cost optimization.

Common Tools: AWS, Azure, Google Cloud consoles, Terraform, and cloud monitoring dashboards.

Certifications: AWS Certified Cloud Practitioner and Microsoft Azure Fundamentals (AZ-900).

6.5 How These Roles Work Together

These four roles work together to keep the company's IT systems running. The Helpdesk Technician handles front-line employee issues and escalates more complex problems. Network-related issues are handled by the Network Administrator, while the Linux System Administrator maintains the Ubuntu server, internal applications, file sharing, and backups. As the company adopts more cloud services, the Cloud Administrator manages cloud resources and disaster recovery. Together, these roles cover user support, network infrastructure, server administration, and cloud operations.

PART 7 – Infrastructure Recommendations

7.1 Internet Provider

A business-grade fiber connection with at least 300 Mbps symmetrical speed is recommended, together with a secondary 5G/LTE connection for failover. This prevents Internet connectivity from becoming a single point of failure.

7.2 Server Specifications

A single rack-mountable or tower server is sufficient for 20 employees. Recommended specifications include an 8-core or better processor, 32–64 GB RAM, RAID-1 or RAID-5 storage, and dual power supplies if the budget allows. Ubuntu Server LTS is recommended to reduce licensing costs and provide long-term support.

7.3 Backup Strategy

A 3-2-1 backup strategy should be used: at least three copies of important data, stored on two different types of media, with one copy kept offsite. Recommended practice includes nightly server-to-NAS backups, weekly external drive rotation, and monthly offsite or cloud backups. Restoration tests should be performed every quarter.

7.4 Security Recommendations

- Segment the network into VLANs per department to reduce lateral movement after a workstation compromise.
- Enable firewall intrusion prevention and restrict inbound access to only required services.
- Require VPN for remote access to internal systems.
- Apply the principle of least privilege to file and system access.
- Keep operating systems and applications patched on a defined schedule.

7.5 Antivirus

Microsoft Defender, included with Windows 11 Pro, provides a suitable baseline for a small company. As the company grows, a managed endpoint detection and response (EDR) solution can be introduced for centralized monitoring.

7.6 Password Policy

- Use passwords with at least 12 characters, including letters, numbers, and symbols.

- Require multi-factor authentication (MFA) for email, VPN, and systems containing client or financial information.
- Rotate administrative passwords every 90 days; standard accounts may follow a longer interval when MFA is enabled.
- Do not reuse any of the previous five passwords.
- Use a company-provided password manager for shared or service accounts.

7.7 Expansion Plan

As the company grows beyond 20 employees, infrastructure should scale in phases. First, additional access points and switch capacity should be added as more users and network drops are required. Next, the server can be virtualized to host more internal services. VLANs and a dedicated guest Wi-Fi network should then be introduced. A hybrid cloud model can be evaluated for cloud backups or disaster recovery. Finally, IT governance should be formalized through documented change-management and employee onboarding/offboarding procedures.

PART 8 – Personal Reflection

Working on this Week 2 portfolio project helped me understand what a System Administrator actually needs to plan before setting up a company network. Before this activity, I mostly thought about computers and servers separately. This project showed me that the hardware, software, network, security, and backups all need to work together.

One of the main things I learned is that the number of employees and the type of work they do affect the equipment that the company needs. For example, HR and Finance can use desktop computers because they usually work in the office, while IT and Sales can use laptops because they may need to move around or work outside the office.

The network diagram was the hardest part for me. I had to understand how the modem, router, firewall, switch, server, access points, and computers connect with each other. It was easier to list the equipment than to show the actual flow of the network.

I also learned that planning can help avoid problems and extra costs. It is better to find a problem while planning than after the equipment has already been bought and installed. I also learned that a System Administrator should be able to explain why a certain device or setup is needed.

Overall, this project helped me practice the main steps of infrastructure planning: identifying business needs, selecting hardware and software, designing the network, identifying system administration roles, recommending security and backup practices, and documenting the plan professionally. These skills will help me become a more prepared and responsible System Administrator.